Intro and First Day Stuff

Lecture 1 - CMSE 381

Prof. Elizabeth Munch

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Mon, Aug 28, 2023

People in this lecture

Dr. Munch (she/her)
Depts of CMSE and Math

Rachel Roca (she/they)
Graduate Student, CMSE, MSU

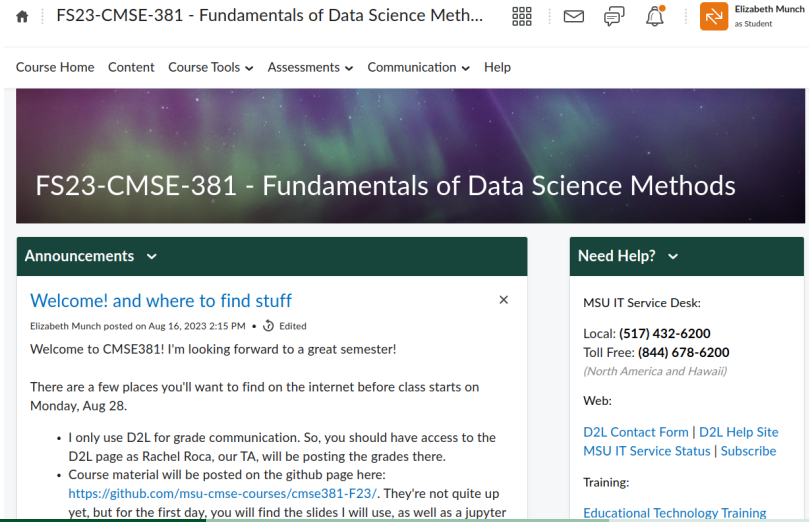
What is this course about?

Topics:

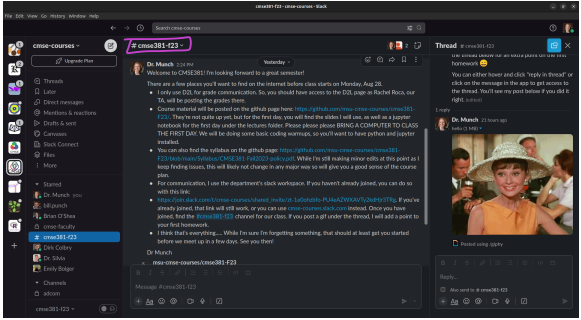
- Fundamental concepts of data science
- Regression
- Classification
- Dimension reduction
- Resampling methods
- Tree-based methods, etc.

D2L and where to find grades

https://d2l.msu.edu/d2l/home/1811231

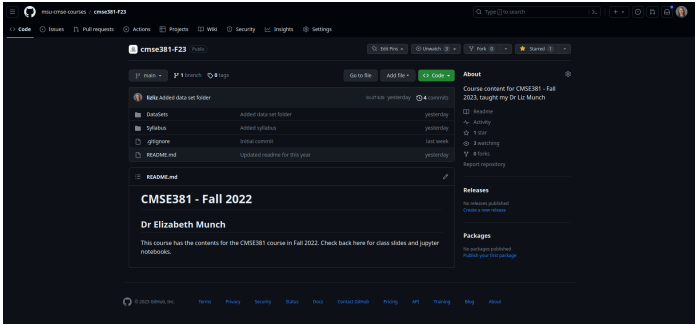


Slack and where to find announcements/ask questions



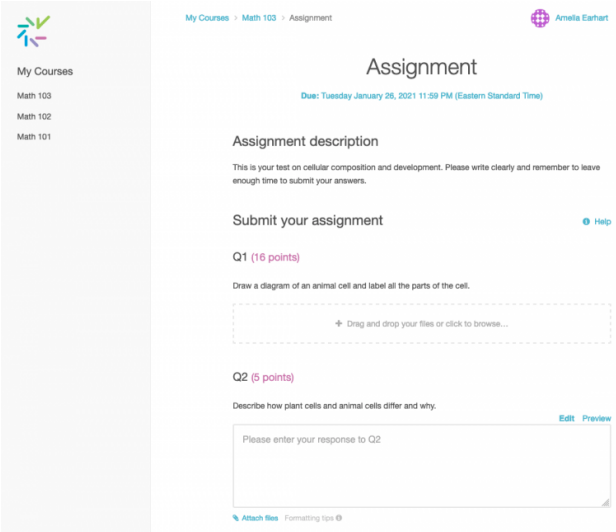
Github and where to find slides and jupyter notebooks

https://github.com/msu-cmse-courses/cmse381-F23/



Crowdmark and where to submit homework

No URL: You will get an automated email from the system (I think.....?)



Office hours

Zoom link: https://bit.ly/3FTuRqG

Dr. Munch

Wednesdays and Thursdays
Both 11am - Noon

Zoom & EGR 1511

Rachel Roca

Tuesdays 3:30 - 5pm,
and Fridays 11:30 - 1pm

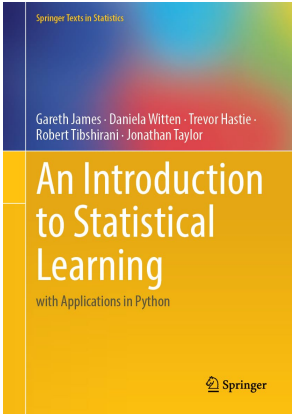
Zoom & EGR (Room ~~TBD~~)

EGR1508A

Textbook

Free download

https://www.statlearning.com/



Class Structure

- Class is a combination of lecture time, and group work/coding time.
  - Bring computer every day
  - Jupyter notebooks
  - Python
- Once a week, there will be a short check-in quiz. This will be basic content realted to lectures since the last class. Possible questions include checking on definitions, or basic understanding of major ideas.
  - 10 points per quiz
  - Drop two lowest grades

Class Structure Pt 2

- Homeworks due once a week, midnight of the day marked in the schedule.
  - 20 points per homework
  - Drop two lowest grades
  - Sliding scale:
    - 24 hours late: 5% penalty.
    - 48 hours late: 15% penalty.
    - >48 hours: No late work accepted.
- Three Midterms
  - See schedule for dates
  - 100 points each
  - Not cumulative
- One Project
  - Analyze dataset using tools in class, submit written report
  - 100 points
  - Due at the end of the semester

Approximate schedule

Up to date version: https://tinyurl.com/CMSE381-FS23-Schedule

Lec #	Date			Reading
1	Mon	Aug 28	Intro / First day stuff / Python Review Pt 1	1
2	Wed	Aug 30	What is statistical learning?	2.1
	Fri	Sep 1	Assessing Model Accuracy	2.2.1, 2.2.2
3	Mon	Sep 4	No class - Labor day	
4	Wed	Sep 6	Linear Regression	3.1
5	Fri	Sep 8	More Linear Regression	3.1/3.2
6	Mon	Sep 11	Even more linear regression	3.2.2
7	Wed	Sep 13	Probably more linear regression	3.3
8	Fri	Sep 15	Intro to classification, Logisitic Regression	2.2.3, 4.1, 4.2, 4.3
9	Mon	Sep 18	More logistic regression	
10	Wed	Sep 20	Multiple Logistic Regression / Multinomial Logistic Regression	
11	Fri	Sep 22	Overflow/Project day?	
	Mon	Sep 25	***Review***	
	Wed	Sep 27	**Midterm #1**	
	Fri	Sep 29	No class - Dr Munch out of town	
12	Mon	Oct 2	Leave one out CV	5.1.1, 5.1.2
13	Wed	Oct 4	k-fold CV	5.1.3
14	Fri	Oct 6	More k-fold CV,	5.1.4-5
15	Mon	Oct 9	k-fold CV for classification	5.1.5
16	Wed	Oct 11	Resampling methods: Bootstrap	5.2
17	Fri	Oct 13	Subset selection	6.1
18	Mon	Oct 16	Shrinkage: Ridge	6.2.1
19	Wed	Oct 18	Shrinkage: Lasso	6.2.2
	Fri	Oct 20	Review	

Lec #	Date			Reading
	Mon	Oct 23	No class - Fall break	
	Wed	Oct 25	Midterm #2	
20	Fri	Oct 27	Dimension Reduction	6.3
21	Mon	Oct 30	More dimension reduction; High dimensions	6.4
22	Wed	Nov 1	Polynomial & Step Functions	7.1,7.2
23	Fri	Nov 3	Step Functions	7.2
24	Mon	Nov 6	Basis functions, Regression Splines	7.3,7.4
25	Wed	Nov 8	Decision Trees	8.1
26	Fri	Nov 10	Random Forests	8.2.1, 8.2.2
27	Mon	Nov 13	Maximal Margin Classifier	9.1
28	Wed	Nov 15	SVC	9.2
29	Fri	Nov 17	SVM	9.3, 9.4
30	Mon	Nov 20	Single layer NN	10.1
31	Wed	Nov 22	Overflow/project day?	
	Fri	Nov 24	No class - Thanksgiving	
	Mon	Nov 27	**Review**	
	Wed	Nov 29	**Midterm #3**	
32	Fri	Dec 1	Multi Layer NN	10.2
33	Mon	Dec 4	CNN	10.3
34	Wed	Dec 6	Unsupervised Learning & Clustering	12.1, 12.4
35	Fri	Dec 8	Overflow/Project day?	
			No final exam	

Grade distribution

	Estimated Points
Homeworks	(10 homeworks - 2 lowest grades) $\times$ 20 points $=$ 160
Quizzes	(12 Quizzes - 2 lowest grades) $\times$ 10 points $=$ 100
Midterm	(3 Midterms) $\times$ 100 $=$ 300
Final Project	100
TOTAL:	660

Section 1

Intro to class

What is Statistical Learning?

Statistical Learning

- Subfield of statistics
- Emphasizes models and their interpretability, precision, and uncertainty

Machine Learning

- Machine learning has a greater emphasis on large scale applications and prediction accuracy.

Very blurred distinction at this point....

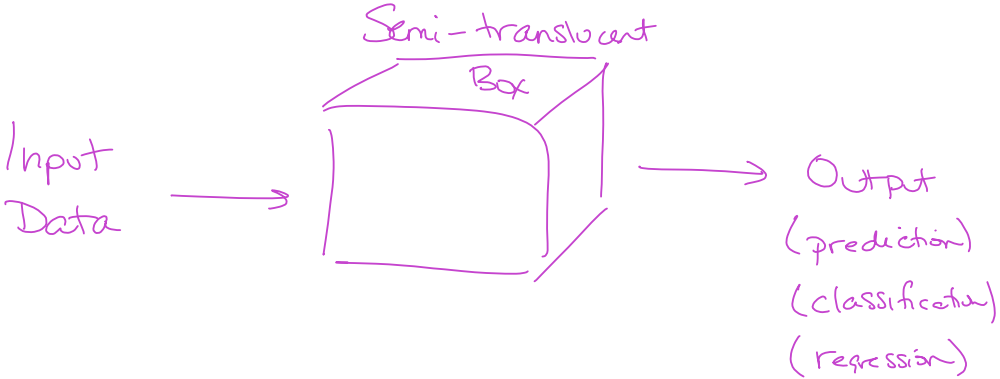
Why should you care?

Data is cheap (or even free), learning how to analyze data is critical.

- Web data, e-commerce (Amazon, JD, Alibaba)
- Car sales (Tesla, Ford, and GM)
- Sports team (MSU, Lions, etc)
- Politics and government

Learning Tools as Black Boxes

- Need to know what tool to use
- Need to know how to interpret output of the tool
- Don't need to rebuild the entire box from scratch



Example: Email spam

	george	you	your	hp	free	hpl	!	our	re	edu	remove
spam	0.00	2.26	1.38	0.02	0.52	0.01	0.51	0.51	0.13	0.01	0.28
email	1.27	1.27	0.44	0.90	0.07	0.43	0.11	0.18	0.42	0.29	0.01

if (%george < 0.6) & (%you > 1.5) then spam
else email.

if $(0.2 \cdot$ %you $- 0.3 \cdot$ %george$) > 0$ then spam
else email.

Take input:
email
Chop up into words

Output
Probability of being Spam

Supervised learning

- Outcome measurement Y (also called dependent variable, response, target, label).
- Vector of p predictor measurements X (also called inputs, regressors, covariates, features, independent variables).

  $X_1, X_2, \ldots, X_p$
- In the regression problem, Y is quantitative (e.g price, blood pressure).
- In the classification problem, Y takes values in a finite, unordered set (survived/died, digit 0-9, cancer class of tissue sample).

  $\mathbb{R}$-value, real number

Unsupervised learning

$\sim Y$ $\sim X_1, \ldots, X_p$

- No outcome variable, just a set of predictors (features) measured on a set of samples.
- Objective is fuzzier: find groups of samples that behave similarly, find features that behave similarly, find linear combinations of features with the most variation.
- Difficult to know how well you are are doing.
- Different from supervised learning but can be useful as a pre-processing step for supervised learning.

dimension reduction

Generative AI discussion

Definition via Wikipedia:

> *Generative artificial intelligence (AI) is artificial intelligence capable of generating text, images, or other media, using generative models. Generative AI models learn the patterns and structure of their input training data and then generate new data that has similar characteristics.*

Examples:

- ChatGPT
- Bard
- DALL-E

- Get in a group of about 4.
- Open this google doc (MSU Login required): tinyurl.com/CMSE381-genAI
- In your group, brainstorm cases where someone might use generative AI in the context of our class.
- Once you have added a few, start adding arguments for or against whether we should allow the use of that context in class.

Section 2

Python Review Lab: Pt 1

Plan for the lab

- Find a group of 4 or so.
- Clone the class repository (or download the jupyter notebook and the csv file from github)
- Get started!

Using git

- `git clone git@github.com:msu-cmse-courses/cmse381-F23.git`
- from inside the folder you just made, run `git pull` any time you want to download new content

Next time

- Weds: What is statistical learning?
- Homework due Friday
- Quiz sometime this week
- Office hours:
  - Dr. Munch: Weds and Thurs 11-12 (At least this week, subject to change)
  - Rachel Roca: Tues 3:30 - 5pm and Fri 11:30 - 1pm

HW

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